

What is a Derivative

(with Applications to Data Science and Physics)

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1 Derivative of a Function, $f : R \rightarrow R$

The derivative of a function, $f : R \rightarrow R$ is a linear approximation to a function. By placing a straight-edge up against the graph of a function at a point we get not just any approximation but the “best” one. Meaning, we could do no better with any other linear approximation. So, the error from this approximation should be higher order than linear. More specifically, we mean:

$$f(x+h) = f(x) + Df(x)(h) + o(h) \quad (1)$$

Here, $Df(x)$ is the linear function that is this best approximation of f at x and $o(h)$ is a higher order error (h^2 , for example) in the sense that $\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$.

Although this geometric concept has been tied to an algebraic equation, it is not clear how to compute the derivative, $Df(x)$ for every x . This seems like a great deal of work and when done we have to produce a function for every x – a relatively complicated thing. It turns out that there is a *representation* result that says that all linear functions, $L, L : R \rightarrow R$ can be mapped to a scalar value. That is, in some sense linear functions are no more complicated than a number. This makes sense as a function that passes through the origin and is a straight line (a linear function) can be completely described by its slope. Specifically, the representation result says that for each linear function, L , there is a scalar, a_L , so that¹

$$L(h) = a_L h \quad (2)$$

Again, this says nothing more than a linear function, L , can be completely determined by its slope, a_L .

For the linear function $Df(x)$ the associated scalar value we label $f'(x)$ and call it –as an abuse of language – the derivative of f at x . To compute this scalar value we use the definition, (1), at a point x :²

$$f'(x)h = Df(x)(h) = f(x+h) - f(x) + o(h) \quad (3)$$

¹We label the scalar with the index L to indicate the dependency on L .

²A convention used here and throughout: $o(h)$ denotes a *generic* higher-order term, possibly a different function from line to line; in particular $-o(h)$ is again such a term, so moving it across an equals sign is harmless.

So that³

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} + \frac{o(h)}{h} \right) \\&= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + 0 \\&= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}\end{aligned}\tag{4}$$

Although this seems like a great deal of work – as for a given function we need to compute this limit for each value of x in its domain – it turns out that for many functions there is a *formula* that works for the given function *for all* points in its domain. For example, the derivative of the function $f(x) = x^2$ at any point can be computed from the formula: $f'(x) = 2x$. In order to avoid the limit process one can build a *calculus* of formulas in the following way:

1. Find formulas for a base collection of functions: power functions, trig, and log functions, etc.
2. Create formulas for computing the derivative of functions formed in certain composite ways: sums, products, ratios, powers, and composition of functions.

For instance, we can build, with the composite operations of sums and products, polynomials from power functions. Consequently, we can easily get formulas for the derivatives of polynomials.

1.1 Application to Optimization

At a local maximum or minimum, x , in the interior of the domain of a function, the tangent line to the graph of the function at x would be flat. That is the linear function, $Df(x)$, should be the zero function; and therefore, $f'(x)$, the corresponding slope of the graph of this linear function should be 0. So, potentially, we have a practical way of finding maxima or minima. This makes sense, to find the maxima or minima of a function over an interval, $[a, b]$, examine all of the local maxima or minima *plus* the values of the function at a and b . This should be a relatively small list. We find the interior candidates – the *critical points* – by solving for all points x such that $f'(x) = 0$.

Example. Find the point x^* which maximizes the function $f(x) = x(12 - 2x)^2$ over the interval $[0, 6]$. We find all points x such that $f'(x) = 0$. It turns out that $f'(x) = 144 - 96x + 12x^2$. Solving for x , $144 - 96x + 12x^2 = 0$ (same as $12 - 8x + x^2 = 0$) we have two solutions: $\{2, 6\}$. The “ x ” values that are candidates which produce the maximum value of the function f are then: $\{0, 2, 6\}$ – the union of the critical points and the boundary points. Now it is a simple matter to walk this list and find an x which produces the maximum value of f . In this case, the “ x ” which produces the maximum value is $x^* = 2$.

We did not show how to compute the derivative, this comes from following the two step plan above. Is the example of any practical value? Well, suppose you wanted to take a square of sheet metal 12 inches on a side, and cut out a square in each corner so that after folding the sides you made a box with the most volume. Then you are really trying to find the length x of the square to cut out so that the volume of the resulting box: $f(x) = x(12 - 2x)^2$ is maximized – the problem above.

³Geometrically, we are approximating the tangent line with lines passing through approximating chords to the function, f .

2 Extension to $f : R^n \rightarrow R$

Imagine standing on a hillside in a thick fog. You cannot see the summit; all you can sense is the tilt of the ground under your feet. Which direction is uphill – and which direction is *steepest*? The altitude is a function of two map coordinates – a function from R^2 to R – and the question is purely local: whatever answer there is must come from the way the hill is tilted right where you stand; that is, from a linear approximation at your feet. To pose such questions – and answer them – we extend the derivative to functions $f : R^n \rightarrow R$. The definition is forced on us by the same idea as before: the derivative is the best linear approximation.

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + Df(\mathbf{x})(\mathbf{h}) + o(\mathbf{h}) \quad (5)$$

Here, $Df(\mathbf{x})$ is a linear function from R^n to R and $o(\mathbf{h})$ is the higher order error term, higher order in the sense that $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{o(\mathbf{h})}{\|\mathbf{h}\|} = 0$.

Again, the function $Df(\mathbf{x})$ is rather abstract and difficult to work with. There is as in the scalar case a *representation* result: Any linear function, L , from R^n to R can be represented by a vector, $\mathbf{a}_L$:

$$L(\mathbf{h}) = \mathbf{a}_L \cdot \mathbf{h} \quad (6)$$

We denote the vector that represents the linear function $Df(\mathbf{x})$ by $\nabla f(\mathbf{x})$. Equation (5) becomes:

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot \mathbf{h} + o(\mathbf{h}) \quad (7)$$

As a calibration, take $n = 1$: the dot product collapses to ordinary multiplication and the representing “vector” collapses to the single number $f'(x)$ – the new machinery reproduces the previous section, which is how one learns to trust it. As in the scalar case, we use equation (5) to compute the components of the vector $\nabla f(\mathbf{x})$ once we choose a coordinate system. To this end let $\{\mathbf{e}_i\}$ be an *orthonormal* basis for R^n – orthonormality is what lets us extract the i^{th} component of a vector by dotting with $\mathbf{e}_i$. That component is

$$\nabla f(\mathbf{x})_i = \nabla f(\mathbf{x}) \cdot \mathbf{e}_i = \lim_{h \rightarrow 0} \left(\frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h} + \frac{o(h\mathbf{e}_i)}{h} \right) \quad (8)$$

If $\tilde{f}$ is the function that takes an n tuple of the components of a vector, $\mathbf{x}$, giving the value of $f(\mathbf{x})$ in such a way that $\tilde{f}(x_1, x_2, \dots, x_n) = f(\mathbf{x})$ and $f(\mathbf{x} + h\mathbf{e}_i) = \tilde{f}(x_1, \dots, x_i + h, \dots, x_n)$, then equation, (8) yields:

$$\begin{aligned} \nabla f(\mathbf{x})_i &= \lim_{h \rightarrow 0} \left(\frac{\tilde{f}(x_1, \dots, x_i + h, \dots, x_n) - \tilde{f}(x_1, \dots, x_i, \dots, x_n)}{h} \right) + \lim_{h \rightarrow 0} \frac{o(h\mathbf{e}_i)}{h} \\ &= \frac{\partial \tilde{f}}{\partial x_i}(x_1, x_2, \dots, x_n) + 0 \\ &= \frac{\partial \tilde{f}}{\partial x_i}(x_1, x_2, \dots, x_n) \end{aligned} \quad (9)$$

Here, we are using the notation $\frac{\partial \tilde{f}}{\partial x_i}$ to “take a derivative” in one direction. That is, we leave all of the other “slots” of $\tilde{f}$ constant and just compute the change in one slot, x_i .⁴

⁴What does x have to do with the i^{th} slot of $\tilde{f}$? Actually nothing, as we can put anything in that slot, $x, y, z, 5$, etc. Keep this in mind, as two sections from now we will see the notation: $\frac{\partial F}{\partial y}$. In this case, it simply means take a derivative in the third slot of the function F .

Note, different choices of coordinate systems will yield different values for the components of ∇f . It turns out that in two dimensions if one uses polar coordinates and the function $\tilde{f}$ is written using these coordinates, the components of the gradient of f in the $(\hat{r}, \hat{\theta})$ basis are $\left[\frac{\partial \tilde{f}}{\partial r}, \frac{1}{r} \frac{\partial \tilde{f}}{\partial \theta}\right]$.

Note. The “local” direction of maximum increase at a point of the $f(\mathbf{x})$ is given by the gradient of f . To see this notice that we can discuss direction of a vector in a unique sense by normalizing it to be of unit length. That is, we can speak of a vector’s direction as its unit vector.

Question. In which direction should we perturb a given point $\mathbf{x}$ so as to increase f as much as possible? That is, find a direction, $\mathbf{w}$ (unit vector), so that for all small enough steps $h > 0$, $f(\mathbf{x} + h\mathbf{w})$ is at least as large as $f(\mathbf{x} + h\mathbf{u})$ for every other unit vector $\mathbf{u}$. We claim that – provided $\nabla f(\mathbf{x}) \neq \mathbf{0}$ – the direction of maximum increase (locally) is $\nabla f(\mathbf{x})$ (normalized).

$$f(\mathbf{x} + h\mathbf{w}) - f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot h\mathbf{w} + o(h\mathbf{w}) \quad (10)$$

But the Cauchy-Schwarz inequality⁵ says: $|\nabla f(\mathbf{x}) \cdot h\mathbf{w}| \leq \|\nabla f(\mathbf{x})\| |h|$. The maximum is only achieved if $\nabla f(\mathbf{x})$ and $\mathbf{w}$ are in the same direction. That is $\mathbf{w} = \frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$.

Therefore, to increase f in the fastest possible way (locally) move in the direction of $\nabla f(\mathbf{x})$; conversely, to decrease f in the fastest possible way (locally) move in the direction of $-\nabla f(\mathbf{x})$. This answers the hillside question that opened the section: the fog is no obstacle, because the question was local all along – the steepest way up is the direction of the gradient, and the steepest way down is its negative.

2.1 Application to Optimization and Data Science

By the same reasoning of the last section in order to find a maximum or minimum one should find all of the places where a function is flat; that is, where the linear approximation is zero. But this is the place where the gradient is zero – the zero linear function is represented by the $\mathbf{0}$ vector. Therefore, find all points $\mathbf{x}$ such that $\nabla f(\mathbf{x}) = \mathbf{0}$. Then compare these points with any boundary points to see which is the largest or smallest.

This might be hard in general – true also in the scalar case from the previous section. How might one go about finding the minimum value of a function, $f(\mathbf{x})$ when solving for $\nabla f(\mathbf{x}) = \mathbf{0}$ is too hard? We know that at any point x , the “local” direction to move to decrease f as much as possible is to move along the direction of the negative gradient. This suggests an iterative scheme – known as *gradient descent* – for finding the minimum.

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 $\mathbf{x}_0 \leftarrow \mathbf{x}^{(0)}$ 
 $n \leftarrow 0$ 
 $converged \leftarrow \text{False}$ 
while not  $converged$  and  $n < N$  do
   $n \leftarrow n + 1$ 
   $\mathbf{x}_n \leftarrow \mathbf{x}_{n-1} - \eta \nabla f(\mathbf{x}_{n-1})$ 
   $converged \leftarrow (\|\mathbf{x}_n - \mathbf{x}_{n-1}\| \leq \epsilon)$ 
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⁵For a proof of the Cauchy-Schwarz inequality – derived from the projection problem that gives birth to the dot product itself – see the companion paper “Where does the Dot Product Come From?”.

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end while

if not converged then
    ERROR: Did not converge within  $N$  steps.
else
     $\mathbf{x}^* \leftarrow \mathbf{x}_n$ 
end if

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This algorithm depends on the parameters:

$\mathbf{x}^{(0)}$: An initial guess;

N : The maximum number of iterations allowed;

ϵ : The error tolerance for convergence;

η : The *learning rate* (step size).

As it is analytically difficult to *solve* $\nabla f(\mathbf{x}) = \mathbf{0}$ for real world problems – the gradient itself is simply evaluated numerically at each step – iterative schemes like the one above are used to find solutions to sophisticated optimization problems. Gradient descent and its variants are also the workhorses used to train deep neural networks.

3 Extension to $f : R^n \rightarrow R^m$

Most models take several inputs to several outputs: a robot arm's n joint angles determine the three coordinates of its hand; a weather model takes today's n readings to tomorrow's m . What plays the role of the derivative for a function $\mathbf{f} : R^n \rightarrow R^m$? The same definition, once more:

$$\mathbf{f}(\mathbf{x} + \mathbf{h}) = \mathbf{f}(\mathbf{x}) + \mathbf{Df}(\mathbf{x})(\mathbf{h}) + \mathbf{o}(\mathbf{h}) \quad (11)$$

Here $\mathbf{Df}(\mathbf{x})$ is a linear function from R^n to R^m and $\mathbf{o}(\mathbf{h})$ is a higher order error term, higher order in the sense that $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|\mathbf{o}(\mathbf{h})\|}{\|\mathbf{h}\|} = 0$.

There is a *representation* result that says that any linear map from R^n to R^m can be represented by a matrix once coordinates have been chosen.

Using (11) one can derive a representation of the derivative using functions commensurate with a choice of basis vectors, $\mathbf{e}_i$ $i \in [1, n]$. Given this choice of basis, let $\tilde{f}_j$ be the coordinate functions for $\mathbf{f}$. We claim without proof that the matrix representing the derivative is the *Jacobian* $[\mathbf{Df}(\mathbf{x})]$, an $m \times n$ matrix whose (i, j) entry (row i , column j) is:

$$[\mathbf{Df}(\mathbf{x})]_{i,j} = \frac{\partial \tilde{f}_i}{\partial x_j} \quad (12)$$

That is, row i corresponds to the i^{th} output component $\tilde{f}_i$, and column j corresponds to the j^{th} input variable x_j . The payoff is the same as at every level: near $\mathbf{x}$ the complicated map $\mathbf{f}$ may be traded for the

linear model $\mathbf{f}(\mathbf{x}) + [\mathbf{Df}(\mathbf{x})] \mathbf{h}$ – the robot arm’s response to small joint adjustments is just a matrix acting on them. And composition of maps becomes multiplication of their Jacobians (the chain rule) – one answer to the question of why matrices multiply the way they do.

Why do we study matrices? One answer is that they represent the linear mappings from $R^n \rightarrow R^m$ and so if we want to use calculus⁶ on the functions from R^n to R^m we need to understand the linear functions, on these spaces and their representations – matrices.

4 Extension to $f : R^\infty \rightarrow R$

A mass hangs from a spring. Between two observed configurations, nature moves it along one particular trajectory out of all the conceivable ones. Which one – and why that one? Physics has a remarkable answer (Hamilton’s principle, taken up at the end of this section): nature’s trajectory renders a certain number, computed from the *whole path*, stationary – a flat spot, just as in the optimization problems of the previous sections. But now the thing being varied is not a number or a vector; it is an entire function. If “set the derivative to zero” is to make sense here, we need a derivative for functions *of functions*.

Consider, then, the general shape of such a path-to-number rule: $J(y) = \int_a^b F(x, y(x), y'(x)) dx$. J is a function from a *function space*, H , to R ; that is, $J : H \rightarrow R$. This is similar to functions, $f : R^n \rightarrow R$, but where n is infinity. One can show that, for instance, the functions $\{\sin(nx)\}_{n=1}^\infty$ are linearly independent; therefore, the space H is infinite dimensional.⁷

We define the derivative of J as before, the best linear approximation:⁸

$$J(y + h) = J(y) + DJ(y)(h) + o(h) \quad (13)$$

Here $DJ(y) : H \rightarrow R$; that is, it is a linear function from function space to R and $o(h)$ is a higher order error term that takes a function, h to a number such that $\lim_{h \rightarrow 0} \frac{o(h)}{\|h\|} = 0$. However, it is not clear what the norm of the function h is. One can define an inner product (generalization of a dot product) on the space of functions which is a generalization of the dot product for vectors in a finite dimensional space. In turn, this inner product defines a norm on the space of functions, H . We define the following inner product function, $\langle \cdot, \cdot \rangle$ by:⁹

$$\langle f, g \rangle \equiv \int_a^b f(x)g(x) dx \quad (14)$$

From this, just as with dot products, we can define the norm, or length of a function, using this inner

⁶We can use calculus to *locally* approximate a given non-linear function, which can be useful in itself. We also know from the last section that we can iterate on the derivative and solve/understand *global* issues of such a non-linear function.

⁷The notation $\{\sin(nx)\}_{n=1}^\infty$ is sloppy, we should instead write: $\{f_n\}_{n=1}^\infty$ where f_n is the function defined by $f_n(x) = \sin(nx)$.

⁸We write $o(h)$ as the left over higher order terms and do not create new function names, even though from equation to equation, the $o(h)$ functions may change.

⁹One caveat inherited from the integral: a function that is zero except at a few isolated points has $\langle f, f \rangle = 0$ without being the zero function. For the *continuous* functions we work with here this cannot happen, so the inner product is honest; the general remedy – identify functions that agree “almost everywhere” – and the geometry of this inner product are taken up in the companion paper “Where does the Dot Product Come From?”.

product:

$$\|f\| = \sqrt{\langle f, f \rangle} \quad (15)$$

As a brief aside we discuss the inner product.

This definition of an inner product seems strange and seems very much removed from the dot product in R^n . We now show that it follows by looking at finite dimensional approximations to H . One way to get a finite version of a continuous function, f , on the interval $[a, b]$ is to sample it at, say, n discrete points on $[a, b]$, $\{x_i\}_{i=1}^n$, which are evenly spaced points over $[a, b]$. We can think of the following vectors as approximations to f and g :

$$\begin{aligned} \tilde{f} &= [f(x_1), f(x_2), \dots, f(x_n)] \\ \tilde{g} &= [g(x_1), g(x_2), \dots, g(x_n)] \end{aligned}$$

The natural inner product of these vectors would be the usual dot product. As we increase the number of points the approximations get closer to the original continuous functions. However, at the same time, due to the increase in the number of points, the induced norm of the approximations gets larger and larger. We would like an inner product that converges to something finite as the approximations converge to the original function. One way to do this is to normalize these dot products so that they would remain finite. The simplest thing to do is to divide by the number of points in the discretization. That is, we define an inner product $\langle\langle \cdot, \cdot \rangle\rangle$ for these vectors as:

$$\langle\langle \tilde{f}, \tilde{g} \rangle\rangle \equiv \frac{1}{n} \sum_{i=1}^n f(x_i)g(x_i) \quad (16)$$

Taking the limit of the above we can define an inner product for the original functions f and g . That is, for our function space, define an inner product, $\langle\langle \cdot, \cdot \rangle\rangle$ by:

$$\begin{aligned} \langle\langle f, g \rangle\rangle &\equiv \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f(x_i)g(x_i) \\ &= \frac{1}{b-a} \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i)g(x_i) \frac{b-a}{n} \\ &= \frac{1}{b-a} \int_a^b f(x)g(x) dx \end{aligned}$$

Now, equation (14) should make sense as an analog of a dot product for vectors. It is a different inner product than the one we just defined – but differs only by a constant multiple.

Going back to the problem at hand we note that there is a *representation* result for linear functions on our function space, H – technically, for a Hilbert space. All linear functions¹⁰, also called functionals, which map the function space to R can be represented by a function in the function space – a generalization of R^n . In R^n we found functionals could be represented by a *vector*. Analogously, for every linear functional, L , in our function space there is a *function*, a_L , such that:

$$L(f) = \langle a_L, f \rangle \quad (17)$$

¹⁰There are some details we won't bother with.

We denote by $\nabla J(y)$ the function that represents the derivative, $DJ(y)$. That is

$$DJ(y)(h) = \langle \nabla J(y), h \rangle \quad (18)$$

Remember, the representation, $\nabla J(y)$, gives us a concrete thing with which we can compute. The derivative is somewhat abstract, it is the representative that really gives us something to work with in terms of computations. This was the case with the derivative of a scalar function as well. There, we represented the derivative, a linear function, as a scalar – the slope of the linear function. The slope, a number, is much easier to use in computations.

We now try to compute $\nabla J(y)$. To recap, we know

$$DJ(y)(h) = J(y+h) - J(y) + o(h) \quad (19)$$

Since $DJ(y)(h) = \langle \nabla J(y), h \rangle$, we have:

$$\langle \nabla J(y), h \rangle = J(y+h) - J(y) + o(h) \quad (20)$$

To find the function $\nabla J(y)$ we need to re-write the RHS of the last equation so that it becomes:

$$\langle \nabla J(y), h \rangle = \langle w, h \rangle + o(h) \quad (21)$$

Then we will have identified our function $\nabla J(y)$: it will be w . Why? To show this, fix a function h and let t be a scalar scaling parameter. By the linearity of the inner product, equation (21) implies that

$$t \langle \nabla J(y), h \rangle = t \langle w, h \rangle + o(t) \quad (22)$$

Dividing by t and taking $t \rightarrow 0$:

$$\langle \nabla J(y) - w, h \rangle = 0 \quad (23)$$

But this calculation is valid for any function h ; consequently, $\langle \nabla J(y) - w, h \rangle = 0 \quad \forall h$. Letting $h = \nabla J(y) - w$, we see that $\nabla J(y) - w = 0$. That is, $\nabla J(y) \equiv w$.

How do we extract h from inside $J(y+h)$? Answer: use Taylor's expansion on F . Continuing, substituting the definition of J in equation (20) and using Taylor's expansion on F we have:¹¹

$$\begin{aligned} \langle \nabla J(y), h \rangle &= \int_a^b F(x, (y+h)(x), (y'+h')(x)) dx - \int_a^b F(x, y(x), y'(x)) dx + o(h) \\ &= \int_a^b \frac{\partial F}{\partial y}(x, y(x), y'(x)) h(x) dx + \int_a^b \frac{\partial F}{\partial y'}(x, y(x), y'(x)) h'(x) dx + o(h) \end{aligned} \quad (24)$$

Here, we used Taylor's expansion on F :

$$F(x, y+h, y'+h') = F(x, y, y') + \frac{\partial F}{\partial y}(x, y, y')h + \frac{\partial F}{\partial y'}(x, y, y')h' + o(h) + o(h') \quad (25)$$

¹¹This is just the definition of the derivative of F : $F(x+a, y+h, y'+h') - F(x, y, y') = \nabla F(x, y, y') \cdot [a, h, h'] + o([a, h, h'])$. In our case, $a = 0$.

If equation (24) looked instead like

$$\langle \nabla J(y), h \rangle = \int_a^b \frac{\partial F}{\partial y}(x, y(x), y'(x)) h(x) dx + o(h) \quad (26)$$

then since (26) must be true for all h in the function space we must have that $\nabla J(y) = \frac{\partial F}{\partial y}(x, y(x), y'(x))$.

Going back to the equation (24), we need to replace the term involving h' with one involving h . Using integration by parts, we can rewrite the second term in (24):

$$\begin{aligned} \int_a^b \frac{\partial F}{\partial y'}(x, y(x), y'(x)) h'(x) dx &= \left. \frac{\partial F}{\partial y'}(x, y(x), y'(x)) h(x) \right|_{x=a}^{x=b} \\ &\quad - \int_a^b \frac{d}{dx} \left[\frac{\partial F}{\partial y'}(x, y(x), y'(x)) \right] h(x) dx \end{aligned} \quad (27)$$

If we restrict the *perturbations* h to functions with $h(a) = h(b) = 0$ – that is, we hold the endpoint values of y fixed, so that the perturbed path $y + h$ must keep them – then the previous equation becomes:

$$\int_a^b \frac{\partial F}{\partial y'}(x, y(x), y'(x)) h'(x) dx = - \int_a^b \frac{d}{dx} \left[\frac{\partial F}{\partial y'}(x, y(x), y'(x)) \right] h(x) dx \quad (28)$$

With this restriction on h , the representative $\nabla J(y)$ is still uniquely determined. This is the *Fundamental Lemma of the Calculus of Variations*: if a continuous function g satisfies $\int_a^b g(x) h(x) dx = 0$ for every smooth h with $h(a) = h(b) = 0$, then $g \equiv 0$. (Sketch: if $g(x_0) \neq 0$, take h to be a small non-negative “bump” concentrated near x_0 , where g keeps one sign by continuity; then $\int_a^b g h dx \neq 0$ – a contradiction. Note that the earlier uniqueness argument – setting $h = \nabla J(y) - w$ – is no longer available here, since that particular h need not vanish at the endpoints.) With this choice of h , equation (24) becomes:

$$\begin{aligned} \langle \nabla J(y), h \rangle &= \int_a^b F(x, (y + h)(x), (y' + h')(x)) dx - \int_a^b F(x, y(x), y'(x)) dx + o(h) \\ &= \int_a^b \frac{\partial F}{\partial y}(x, y(x), y'(x)) h(x) dx + \int_a^b \frac{\partial F}{\partial y'}(x, y(x), y'(x)) h'(x) dx + o(h) \\ &= \int_a^b \left[\frac{\partial F}{\partial y}(x, y(x), y'(x)) - \frac{d}{dx} \frac{\partial F}{\partial y'}(x, y(x), y'(x)) \right] h(x) dx + o(h) \end{aligned} \quad (29)$$

Using (21) we find that

$$(\nabla J(y))(x) = \frac{\partial F}{\partial y}(x, y(x), y'(x)) - \frac{d}{dx} \left[\frac{\partial F}{\partial y'}(x, y(x), y'(x)) \right] \quad (30)$$

We ignored a few technical details – when h is small in our distance measure is h' small? However, the point is that as far away as we are from the scalar case, the essence of the calculations are the same – identify the linear thing acting on a small change to the input, h , and show that what remains are “higher order” terms.

4.1 Application to Physics

Certain systems in physics have the property that the dynamics of the system are governed by minimizing (or rendering stationary – hitting a flat spot) a certain functional, called the *action*. This is *Hamilton's principle of stationary action*. At a stationary trajectory the derivative of the action should be “zero”; meaning, in this case, the zero function.¹² From this condition we can determine the dynamics of the system. That is, for certain physical systems, the path trajectory of a particle, $y(x)$, is determined by finding the function y where the gradient, $\nabla J(y)$, is the zero function. Using (30), $\nabla J(y) \equiv 0$ means:

$$\frac{\partial F}{\partial y}(x, y(x), y'(x)) - \frac{d}{dx} \left[\frac{\partial F}{\partial y'}(x, y(x), y'(x)) \right] = 0 \quad (31)$$

Equation (31) is the celebrated *Euler–Lagrange equation* of the calculus of variations.

Example.

We can now answer the question that opened this section: which trajectory does the mass on the spring actually follow? The dynamics of a mass is determined by finding a trajectory at which the *action* J is stationary – a place where the derivative of the action is the zero function. The integrand F of the action is called the *Lagrangian*; for a mechanical system it is the kinetic energy minus the potential energy. For a mass attached to a spring (with y measuring displacement from the equilibrium position, so any constant gravitational potential drops out and the Lagrangian reduces to kinetic minus elastic potential energy) the action is¹³

$$J(y) = \int_0^T \overbrace{\frac{1}{2} m y'(t)^2}^{\text{Kinetic Energy}} - \overbrace{\frac{1}{2} k y(t)^2}^{\text{Potential Energy}} dt \quad (32)$$

In this case, the Lagrangian is $F(t, y, y') = \frac{1}{2} m y'^2 - \frac{1}{2} k y^2$. Therefore, we need to find a function, y , such that $(\nabla J(y))(t) = 0 \quad \forall t \in [0, T]$. This equation becomes (using the dummy variable t):

$$\frac{\partial F}{\partial y}(t, y(t), y'(t)) - \frac{d}{dt} \left[\frac{\partial F}{\partial y'}(t, y(t), y'(t)) \right] = 0 \quad (33)$$

The partial derivatives of F are computed as:

$$\frac{\partial F}{\partial y} = -ky \quad (34)$$

$$\frac{\partial F}{\partial y'} = m y' \quad (35)$$

Substituting into (33) we have:

$$-k y(t) - \frac{d}{dt} [m y'(t)] = 0 \quad (36)$$

¹²The zero function is the function which sends all inputs to the value 0.

¹³We replace the “dummy” variable x in our equations with t .

Or,

$$my''(t) + ky(t) = 0 \quad (37)$$

Pause on what just happened: this is exactly Newton's second law, $F = ma$, for the Hooke spring force $F = -ky$. The new machine – action, representative, Euler–Lagrange – reproduces the answer the reader already owns, which is how one learns to trust a new tool.

This is a second order linear differential equation. The general form of a solution to this differential equation is:

$$y(t) = a_1 \cos(\sqrt{k/m}t) + a_2 \sin(\sqrt{k/m}t) \quad (38)$$

The variational problem above held the positions at times 0 and T fixed; but once the differential equation is in hand, any equivalent data determine the two constants – given, say, the initial position and velocity we can determine a_1 and a_2 .

5 The Same Game on Four Boards

This paper has been one game played four times. At every level the two moves were the same: *define* the derivative as the best linear approximation, then *trade* the abstract linear map for a concrete representative you can compute with. Only the board changed:

Function	Representative	Pairing	“Set it to zero”
$f : R \rightarrow R$	slope $f'(x)$	multiplication	critical points (the box)
$f : R^n \rightarrow R$	gradient $\nabla f(\mathbf{x})$	dot product	gradient descent
$\mathbf{f} : R^n \rightarrow R^m$	Jacobian $[\mathbf{D}\mathbf{f}(\mathbf{x})]$	matrix product	(local linear model)
$J : H \rightarrow R$	function $\nabla J(y)$	$\langle f, g \rangle = \int_a^b fg \, dx$	Euler–Lagrange

What “multiply” means changes from row to row – a product of numbers, a dot product, a matrix acting on a vector, an integral – but the strategy never does. A reader who owns the first row already owns the last; the only new work at each level was finding the right representation.

This representative-of-a-linear-functional theme is one strand of a larger web. The dot product and its inner product generalizations – including the Cauchy–Schwarz inequality used in section 2 and the integral inner product built in section 4 – are derived from a single projection problem in the companion paper “Where does the Dot Product Come From?”. And the “best approximation” idea that defined the derivative reappears, in the same integral inner product, as the best guess of probability theory in “Measure Theoretic Conditional Expectation in an Elementary Setting”. Same game; bigger boards still.

Further reading. For the derivative as a linear map, see M. Spivak, *Calculus on Manifolds*. For the calculus of variations and the Fundamental Lemma, see I. M. Gelfand and S. V. Fomin, *Calculus of Variations*. For the action principle in mechanics, see H. Goldstein, *Classical Mechanics*.